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Decentralized Coordination Model For UAV Swarm Based On Federated Deep Learning

With the rapid development and increasing use of UAV in various spheres of life, they still face a number of challenges, such as security and privacy issues, as well as the likelihood of operations failing as a result of poor or interrupted connection.

Problem Statement

UAV Swarms systems suffer from major security and privacy challenges, as well as high dependence on connectivity, which can lead to process failures in some cases. They are also at risk of penetration.

Methodology

The proposed model is built on three layers:

- **Communication network layer.**
- **Blockchain layar.**
- **Control and decision layer.**

Simplified Model Architecture

Connection loss

Checks connected UAV

Leaders are selected according to the highest score

1. **Divides followers into leaders by distance, power and connection quality.**
2. **Followers collect data and send it to the leader as a model.**
3. **The leader shares it with other leaders and they agree on the decision.**
4. **Leaders send decisions to followers as a model.**

Periodically, connected drones are re-validated, and leader–follower roles are reassigned as needed.

Objectives

- Develop a model based on Federated Learning to enhance security.
- Integrate Blockchain technology to enhance security and ensure the reliability of Model exchange.
- Improve the level of privacy in UAV Swarms environments.
- Reduce dependence on ground control stations.
- Enable the swarm to coordinate and complete tasks independently.
- Reduce the likelihood of operations failures.
- Enable the swarm to navigate and complete tasks when GPS is lost.

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